

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from mpl_toolkits.mplot3d import Axes3D
```

```
df=pd.read_csv("/content/drive/MyDrive/Real estate.csv")
```

```
df.shape
```

```
(414, 8)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 414 entries, 0 to 413
Data columns (total 8 columns):
#   Column                                Non-Null Count  Dtype
---  -
0    No                                     414 non-null    int64
1    date                                 414 non-null    float64
2    house age                           414 non-null    float64
3    distance to the nearest station     414 non-null    float64
4    number of convenience stores        414 non-null    int64
5    latitude                             414 non-null    float64
6    longitude                           414 non-null    float64
7    house price of unit area            414 non-null    float64
dtypes: float64(6), int64(2)
memory usage: 26.0 KB
```

```
df.columns
```

```
Index(['No', 'date', 'house age', 'distance to the nearest station',
      'number of convenience stores', 'latitude', 'longitude',
      'house price of unit area'],
      dtype='object')
```

```
#Define features (X) and target (y)
X = df[['house age', 'distance to the nearest station',
      'number of convenience stores', 'latitude', 'longitude']] # independent variables
y = df['house price of unit area'] # dependent variable
```

```
X.head()
```

	house age	distance to the nearest station	number of convenience stores	latitude	longitude
0	32.0	84.87882	10	24.98298	121.54024
1	19.5	306.59470	9	24.98034	121.53951
2	13.3	561.98450	5	24.98746	121.54391
3	13.3	561.98450	5	24.98746	121.54391
4	5.0	390.56840	5	24.97937	121.54245

```
#Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
#Train Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)
```

```
LinearRegression
```

```
#Predictions
y_pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_squared_error, r2_score
```

```
#Evaluation
print("Coefficients (slopes):", model.coef_)
```

```
print("Intercept:", model.intercept_)
print("Mean Squared Error (MSE):", mean_squared_error(y_test, y_pred))
print("R² Score:", r2_score(y_test, y_pred))
```

```
Coefficients (slopes): [-2.70593236e-01 -4.55249601e-03  1.10512079e+00  2.36092831e+02
-2.39036942e+01]
Intercept: -2946.6588590244446
Mean Squared Error (MSE): 54.58094520086248
R² Score: 0.6746481382828156
```

```
import pickle
```

```
pickle.dump(model, open('REPPModel.pkl', 'wb'))
```

```
from flask import Flask, render_template, request
import pickle
import numpy as np

app = Flask(__name__)

# Load the trained model
model = pickle.load(open("sleepnight.pkl", "rb"))

@app.route('/')
def home():
    return render_template("index.html")

@app.route('/predict', methods=['POST'])
def predict():

    # Get input values from HTML form
    body_weight = float(request.form['body_weight'])

    brain_weight = float(request.form['brain_weight'])

    max_life_span = float(request.form['max_life_span'])

    gestation_time = float(request.form['gestation_time'])

    predation_index = float(request.form['predation_index'])

    sleep_exposure_index = float(
        request.form['sleep_exposure_index']
    )

    danger_index = float(
        request.form['danger_index']
    )

    # Arrange inputs in EXACTLY the same order
    # used while training the model
    input_data = np.array([[
        body_weight,
        brain_weight,
        max_life_span,
        gestation_time,
        predation_index,
        sleep_exposure_index,
        danger_index
    ]])

    # Predict total sleep
    prediction = model.predict(input_data)

    # Display prediction
    prediction_text = (
        f"Predicted Total Sleep : "
        f"{prediction[0]:.2f} hours"
    )

    return render_template(
        "index.html",
        prediction_text=prediction_text
    )
```

```
if __name__ == "__main__":  
    app.run(debug=True)
```

```
<!DOCTYPE html>  
<html lang="en">  
  
<head>  
  
    <meta charset="UTF-8">  
  
    <meta name="viewport"  
        content="width=device-width, initial-scale=1.0">  
  
    <title>Real Estate Price Prediction</title>  
  
    <style>  
  
        * {  
            margin: 0;  
            padding: 0;  
            box-sizing: border-box;  
            font-family: Arial, Helvetica, sans-serif;  
        }  
  
        body {  
  
            background: #f2f6fc;  
  
            display: flex;  
  
            justify-content: center;  
  
            align-items: center;  
  
            min-height: 100vh;  
        }  
  
        .container {  
  
            width: 500px;  
  
            background: white;  
  
            padding: 35px;  
  
            border-radius: 15px;  
  
            box-shadow:  
                0px 8px 25px  
                rgba(0, 0, 0, 0.15);  
        }  
  
        h1 {  
  
            text-align: center;  
  
            color: #1565c0;  
  
            margin-bottom: 8px;  
        }  
  
        h3 {  
  
            text-align: center;  
  
            color: #666;  
  
            margin-bottom: 25px;  
  
            font-weight: normal;  
        }  
    </style>  
</head>  
  
<body>  
  
    <div class="container">  
  
        <h1>Real Estate Price Prediction</h1>  
  
        <h3>Welcome to the Real Estate Price Prediction App</h3>  
  
        <div class="form">  
            <input type="text" value="Enter your location here" />  
            <input type="button" value="Predict Price" />  
        </div>  
    </div>  
</body>  
</html>
```

```
label {  
  
    display: block;  
  
    font-weight: bold;  
  
    margin-top: 14px;  
  
    margin-bottom: 6px;  
  
    color: #333;  
  
}  
  
input {  
  
    width: 100%;  
  
    padding: 11px;  
  
    font-size: 16px;  
  
    border: 1px solid #ccc;  
  
    border-radius: 7px;  
  
    outline: none;  
  
}  
  
input:focus {  
  
    border-color: #1565c0;  
  
}  
  
button {  
  
    width: 100%;  
  
    margin-top: 25px;  
  
    padding: 13px;  
  
    background: #1565c0;  
  
    color: white;  
  
    border: none;  
  
    border-radius: 8px;  
  
    font-size: 18px;  
  
    cursor: pointer;  
  
}  
  
button:hover {  
  
    background: #0d47a1;  
  
}  
  
.result {  
  
    margin-top: 25px;  
  
    padding: 18px;  
  
    background: #e8f5e9;  
  
    color: #2e7d32;  
  
    border-radius: 8px;  
  
    text-align: center;
```

```
        font-size: 20px;

        font-weight: bold;

    }

    footer {

        text-align: center;

        margin-top: 20px;

        color: #777;

        font-size: 13px;

    }

</style>

</head>

<body>

<div class="container">

    <h1>Real Estate Price Predictor</h1>

    <h3>Multiple Linear Regression</h3>

    <form action="/predict" method="POST">

        <label>
            House Age
        </label>

        <input
            type="number"
            name="house_age"
            step="0.01"
            placeholder="Example: 13.3"
            required
        >

        <label>
            Distance to Nearest Station
        </label>

        <input
            type="number"
            name="distance"
            step="0.01"
            placeholder="Example: 4082.0"
            required
        >

        <label>
            Number of Convenience Stores
        </label>

        <input
            type="number"
            name="convenience_stores"
            min="0"
            step="1"
            placeholder="Example: 2"
            required
        >

        <label>
            Latitude
        </label>
```

```
<input
  type="number"
  name="latitude"
  step="0.000001"
  placeholder="Example: 24.9830"
  required
>

<label>
  Longitude
</label>

<input
  type="number"
  name="longitude"
  step="0.000001"
  placeholder="Example: 121.5400"
  required
>

<button type="submit">

  Predict House Price

</button>

</form>

{% if prediction_text %}

<div class="result">

  {{ prediction_text }}

</div>

{% endif %}

<footer>

  By Mr. Aditya Tanaji Gawade

</footer>

</div>

</body>

</html>
```

Start coding or [generate](#) with AI.